MILITARY SPECIFICATION

Rack, Electrical Equipment MT-4484/APM

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for intermediate shop use and shall, when utilized with the preferred standard test equipment listed herein, provide the capability for bench testing the electrical assemblies of the RT-1028/APN-202 Receiver Transmitter (See 6.1).

1.2 Classification - The equipment covered by this specification shall consist of the Rack, Electrical Equipment MT-4484/APM.

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-N-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

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MIL-T-21200 Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-461 Electro Magnetic Interference Characteristic, Requirements for Equipment

MIL-STD-462 Electro Magnetic Interference Characteristics, Measurement of

MIL-STD-463 Definitions and Systems of Units, Electromagnetic Interference Technology.

2.1.1 Availability of Documents -

(1) When requesting specifications, standard, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.3.1).

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(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained in Specification MIL-T-21200.

3.3 Design and Construction - The equipment shall conform with all the applicable-requirements of Specification MIL-T-21200 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 38 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 300 hours without the necessity for readjustment of any controls, which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement, due to wear during the life of the equipment and the wearout life of such parts, shall be specified by the contractor.

3.3.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted such that the connecting cables will not interfere with the operating controls. Permanently numbered wires may be used in lieu of color coding.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of Specification MIL-STD-461.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification

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MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{o} \pm 10^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input dc voltage	28 $\pm$0.5 Vdc

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions, as specified in MIL-T-21200 for class 3 equipment.

3.3.9.1 Shock - The equipment shall operate satisfactorily after being subjected to 20g shock pulses per MIL-T-21200.

3.3.10 Warm-up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.3.11 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200. The power required shall be of the following type and shall not exceed the specified amounts.

DC Power, 28 Vdc, 1.5 amperes

3.4 Performance - The performance of the equipment shall be as set forth in paragraph 3.5. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

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3.4.1 Bench Test Parameters - The performance characteristics specified herein for the Rack include the use of the standard and common test equipment specified in 6.8.

3.4.1.1 IF Amplifier Test Parameters - External instrumentation shall provide a 50 to 75 MHz signal (swept cw or pulsed rf). Signal level shall be capable of being varied from 0 to -100 dBm. The following parameters shall be measured with the indicated absolute accuracy.

Parameter	Accuracy
a. IF Amplifier Gain	±2 dB
b. 3 dB Passband	±2 MHz
c. Center Frequency	±1 MHz
d. Threshold Attenuation	±2 dB
e. Video Amplitude Modes	±5%

3.4.1.2 Pulse Assembly Test Parameters - External instrumentation shall provide single or double pulses with the following pulse characteristics:

Rise Time	0.1 usec max.
Fall Time	0.2 usec max.
Width	0.3 to 0.6 usec
Amplitude	0.2 to 5 volts (negative)
Repetition rate	500 to 5000 pps

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Video Amplitudes	±5%
b. Built in test (BIT) functions	visual indication of go-no-go
c. Power line check	±5%
d. Decoding Adjustment	±0.5 usec

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Parameter	Accuracy
e. Decoding test	±0.05 usec

3.4.1.3 Modulator Test Parameters - External instrumentation shall provide an interrogation pulse with the following characteristics:

Rise Time	0.1 usec max.
Fall Time	0.2 usec max.
Width	0.3 to 0.6 usec
Amplitude	0 to 10 volts (positive)
Repetition Rate	0.5 to 15 kHz

The following parameters shall be measured to the indicated absolute accuracy.

Parameter	Accuracy
a. Switch Drive-current Pulse	±3%
b. Video Amplitudes	±10%
c. Decoder Trigger Level	±0.1 volt
d. Minimum Pulse Delay	±0.025 usec.
e. Over-Interrogation	±1 count
f. Encode Pulse Spacing	±1%
g. Recovery Blanking	±0.025 usec.

3.4.1.4 Power Supply Test Parameters - External instrumentation shall measure the following parameters to the indicated absolute accuracy.

Parameter	Accuracy
a. Regulation	±3%
b. DC Voltages	±0.05%
c. AC Voltages	±5%
d. Input Current Drain	±1%

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Parameter	Accuracy
e. Overvoltage Limits	±1%

3.5 Detail Requirements

3.5.1 Rack, Electrical Equipment MT-4484/APM - The Rack shall provide test fixtures to test the four assemblies (listed in 6.1) of the RT-1028/APN-202 Receiver Transmitter. It shall include means for securing the assembly to the test fixture and provide heat sinks where necessary. It shall provide the necessary power, loads, voltage dividers, signal processing circuits, switches, test points, and connectors to facilitate the use of the preferred standard test equipment to test the assemblies for satisfactory electrical performance.

3.5.1.2 Form Factor - The Rack shall be mounted in a carrying case. Maximum dimensions shall not exceed 23.5 inches by 10 inches by 16.5 inches.

3.5.1.3 Weight - The weight of the Rack shall not exceed 38 pounds.

3.5.1.4 Content - The Rack shall contain the following assemblies.

3.5.1.4.1 Internal Power Supply - The Rack shall incorporate a power supply essentially identical to the RT-1028/APN-202 power supply for providing power to the subassemblies under test. With the input voltage set to 28 ±0.5 VDC, the following voltages shall be provided at the maximum ripple indicated when loaded with the normal operating load.

Voltages	Ripple
+105 ±10 VDC	0.2V rms max.
+20 ±0.5 VDC	0.1V rms max.
+10 ±0.5 VDC	0.05V rms max.
+10 (Delayed) +2, -1 VDC	0.05V rms max.
-10 ±0.5 VDC	0.05V rms max.
6.4 ±0.3 VRMS	(not applicable)

3.5.1.4.2 Threshold Detector/Video Assembly - The Rack shall contain an assembly incorporating the following circuits required to test the RT-1028/APN-202 I.F. Amplifier assembly.

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(a) A threshold detector similar to that in the Pulse Assembly of the RT-1028/APN-202 Receiver Transmitter. The detector shall serve as an interface between the video output of the I.F. Amplifier assembly and external monitoring counter. Characteristics of the threshold detector shall be as follows:

Threshold Input Pulse Amplitude . . -025 ±0.1 volt

Input pulse width . 0.4 ±0.1 microsec.

Output Pulse Amplitude . . . . . 2.0 volts min.

(b) A video amplifier circuit to serve as an interface between a probe and an external oscilloscope to permit visual presentation of the bandpass characteristics of the various stages in the I.F. Amplifier assembly. Characteristics of the video amplifier shall be as follows:

Gain . . . . . 14 ±4

3 dB upper cutoff freq. . . . . 13 ±4 KHz

3 dB lower cutoff freq. . . . . Less than 20 Hz

(c) A BIT pulse generator circuit similar to that in the Pulse Assembly of the RT-1028/APN-202. This generator shall be used to trigger the BIT oscillator of the I.F. Amplifier assembly. When the output of the generator is connected to a voltage divider consisting of 27K ohms to +28 VDC and 1800 ohms to ground, the pulse output characteristics shall be as follows:

Pulse Amplitude . . 1.1 ±0.4 volt negative

Pulse Width . . . . 1.0 ±0.3 microsec.

Repetition Rate . . 850 ±250 pps

3.5.1.4.3 IF Amplifier Test Fixture Assembly - A shielded enclosure shall be provided to test the I.F. Amplifier. The enclosure shall include a dummy mixer ahead of the input to the

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I.F. Amplifier under test. Nominal input impedance of the dummy mixer shall be 50 ohms. The nominal output impedance shall be 200 ohms shunted by a 5 pf capacitor.

3.5.1.4.4 Modulator Load - Voltage Divider Assembly - The Rack shall provide an assembly containing a load for the output of the modulator assembly and also include the encoder control and decoder control voltage divider networks as follows:

(a) The modulator load shall simulate the anode voltage-current characteristics of a magnetron wherein load-current flow does not occur below approximately 2200 volts. Provisions shall be made to monitor the load-pulsed current on an oscilloscope. Characteristics of the load shall be as follows:

Peak Pulse Voltage -2700 ±100 volts

Peak Pulse Current 1.45 ±0.15 amperes

(b) The precision encoder and decoder voltage dividers and the associated control switching shall be similar to the functions provided by the system Control Panel, C-8752/APN-202.

3.5.1.4.5 Power Supply Load Assembly - The Rack shall provide an assembly containing loads for the power supply assembly under test. The nominal loads for the various outputs of the power supply shall be as follows:

Output	Load
+110 VDC*	0 (Standby)
	30 ma. (Normal)
	110 ma. (Max.)
+ 20 VDC	91 ma.
+ 10 VDC	147 ma.
+ 10 VDC (Delayed)*	1.2 ma.
- 10 VDC	147 ma.
6.4 VRMS	850 ma.

* NOTE: The +110 VDC load shall be shunted with a 10 uf capacitor and the +10 VDC (Delayed)load shall be shunted with a 6.8 uf capacitor.

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3.5.1.4.6 Overload Protection Assembly - The Rack shall incorporate overload protection circuitry to prevent damage to the internal power supply due to short circuits on any of the outputs. A manual reset shall be provided.

3.5.1.5 Controls - The Rack shall have the following controls:

(a) TEST SPLY PWR - Switch permitting the selection of dc power from the AN/APM-231 or an auxiliary external supply. The following positions are required: 28V, EXT SPLY.

(b) VOLTAGE SELECT - A nine-position rotary switch that operates in conjunction with the MODULE SELECT switch. Permits selection of test voltages to external monitoring voltmeters.

(c) FAULT TEST - A two-position switch used for test of the I-F Amplifier BIT circuits. The following positions are required: ON, OFF.

(d) A two-position switch permitting the selection of the I-F Amplifier video test point or the I-F Amplifier video output. The following positions are required: VIDEO, VIDEO TP.

(e) MODE - A six-position switch for operation of the control inputs to the modulator under test. The following positions are required: OFF, DCDR TRIG, DLY INHB, ACLS TRIG, EXT DATA, DATA INHB.

(f) ENCODE - A six-position switch used for operation of the encoder in the modulator under test. The following positions are required: S, 1, 2, 3, 4, 5.

(h) A two-position switch used to operate the standby control input to the pulse assembly under test. The following positions are required: STBY, NORM.

(i) CODE - A six-position switch used to operate the code control input to the pulse assembly under test. The following positions are required: S, 1, 2, 3, 4, 5.

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(j) A two-position switch used to route the output of the BIT oscillator of the pulse assembly under test. The following positions are required: NORM, FAULT.

(k) POWER - A three-position switch used to route primary power. The following positions are required: INTL SUPLY, OFF/RESET, TEST SPLY.

(l) A two-position switch used to apply a voltage to the over-interrogation blanking input of the pulse assembly under test. The following positions are required: OI BLK, NORM.

(m) LOAD - A three-position switch used to select the loading conditions of the power supply assembly under test. The following positions are required: STBY, NORM, MAX.

(n) MODULE SELECT - An eight-position voltmeter control switch for selecting the module whose voltages are to be measured. Operates in conjunction with the VOLTAGE SELECT switch. The following positions are required: OFF, INTL, PULSE ASSY, PWR SPLY V, TEST PWR SPLY CUR, IF BIAS, MOD TEST, MOD PWR SPLY.

3.5.1.6 Electrical Connections - The Rack shall have the following electrical connections, test points, and indicators:

3.5.1.6.1 Connections

Reference Designation	Receptacle Type	Function
J1	MS3112E-12-10P	+28 VDC power input from the C-6690A/APM Control, Test Harness
J3	M39024/10-01 Tip Jack	Test jack for monitoring 6.3 VRMS output
J4	M39024/10-03 Tip Jack	Test jack for monitoring 6.3 VRMS output
J5	M39024/10-02 Tip Jack	Test jack for monitoring d-c voltage outputs
J6	Same as J5	Test jack for monitoring d-c voltage outputs

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Reference Designation	Receptacle Type	Function
J7	M39012/19-0002 Coax, Series "BNC"	RF input to I.F. Assembly test fixture
J8	M39012/21-0002 Coax, Series "BNC"	Video output from I.F. Amplifier test fixture
J9	Same as J8	Processed video output from threshold detector of I.F. Amplifier test fixture
J10	Same as J8	Output to scope from probe bandpass amplifier on I.F. Amplifier test fixture
J11	Same as J8	Pulse trigger input to modulator assembly test fixture
J12	Same as J8	Monitor jack for pre-trigger output of Modulator assembly test fixture
J13	Same as J8	Monitor jack for BIT pulse generator output of pulse assembly under test
J14	Same as J8	Monitor jack for processed video output of pulse assembly under test
J15	Same as J8	Pulse trigger input to pulse assembly test fixture
J16	Same as J8	Monitor jack for pulse trigger input of J15
J17	Same as J8	Monitor jack for decoder output of pulse assembly under test
J18	Same as J8	Monitor jack for viewing the current pulse of the modulator load of the modulator assembly test fixture
J19	MS3112E-8-4P	DC power input from auxiliary power supply.
J25	Same as J8	Probe jack input to bandpass amplifier of I.F. amplifier test fixture

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Reference Designation	Receptacle Type	Function
J26	Motorola type 28-P19235B001	Assembly test connector. Mates with I.F. amplifier assembly.
J27	Motorola type 28-P19237B001	Assembly test connector. Mates with modulator assembly.
J28	M24308/3-1 9 pin male	Assembly test connector. Mates with modulator assembly through jumper cable.
J29	Motorola type 28-P19232B001	Assembly test connector. Mates with pulse assembly.
J30	Motorola type 28-P19234B001	Assembly test connector. Mates with pulse assembly.
J31	M24308/1-2 15 pin female	Assembly test connector. Mates with pulse assembly through jumper cable.
J32	M24308 3-2 15 pin male	Assembly test connector. Mates with pulse assembly through jumper cable.
J33	M24308/1-1 9 pin female	Assembly test connector. Mates with power supply through jumper cable

3.5.1.6.2 Indicators - Test indicators, not including power on, shall be as follows:

Indicator	Function
FAULT	Indicator the same as the type used in the RT-1028/APN-202 Receiver-Transmitter. Checks pulse assembly fault set and reset circuits.
LAMP DRIVER	28 volt/40 mA. lamps used in checking the lamp driver circuits of the pulse assembly.
OVERLOAD	Indicator used to indicate the overload has been tripped on the internal power supply.

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3.5.2 Cable Assemblies - Cable assemblies shall be provided as follows:

3.5.2.1 Special Purpose Cables -

Cable Designation	Length (inches)	Function
W1	30 ±1	DC power input cable from J5 of Control, Test Harness, C6690A/APM-231
W2	8.5	Assembly test cable. Connects from J28 of the Rack to J1 of the modulator assembly.
W3	10.5	Assembly test cable. Connects from J31 of the Rack to J2 of the pulse assembly.
W4	6.5	Assembly test cable. Connects from J32 of the Rack to XA2 of the pulse assembly.
W5	6	Assembly test cable. Connects from J33 of the Rack to P1 of the power supply assembly.
W6	72	Power cable. Connects from J19 of the Rack to an auxiliary d-c power supply.
W8	48	Monitor cable. Connects from J3 and J4 of the Rack to the RMS voltmeter.

3.5.2.2 Connectors - Connectors for the special purpose cables shall be as follows:

Cable Designation	P1	P2
W1	MS3116F-12-10S	MS3116F-12-10P
W2	M24308/1-1	Motorola type 28-P19197B001

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Cable Designation	P1	P2
W3	M24308/3-2	Motorola Type 28-P19231B001
W4	M24308/1-2	Motorola Type 28-P19236B001
W5	M24308/3-1	Motorola Type 28-P19233B001
W6	MS3116F-8-4S	Strip and tinned ends approximately 0.5 inches
W8	Tip plug, mates with J3 and J4 (two required)	M39012/16-0001

3.5.2.3 Cables, RF - Coaxial cable assemblies shall be as follows:

Cable Designation	Approximate Length (inches)	Function
W7	18	RF input line from attenuators to J7 of the Rack.
W7	18	Connects RF generator to attenuators.

3.5.2.4 Connectors, RF - Connectors for the RF cables shall be as follows:

Cable Designation	P1	P2
W7	M39012/16-0001	Same as P1
W7	M39012/16-0001	Same as P1

3.6 Design Data - No data is required by this specification unless specified in the contract or purchase order (see para. 6.6).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures of 4.4. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include applicable environmental and interference tests in accordance with the requirements of Specifications MIL-T-5422 and MIL-STD-461 to the limits specified in Specification MIL-T-21200 except as modified herein.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the production sample is at the contractor's own risk.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection. For equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-in Test.

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.3.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.3.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0
Next 50	1

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Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in paragraph 4.3.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause the remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.3.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.3.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.3.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.3.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or to require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. Specification MIL-T-18303 shall be used as a guide for preproduction of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary.

4.5 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shal resubmit the equipment for acceptance.

4.6 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to

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comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The intended use of this equipment shall be to facilitate the shop testing in conjunction with the associated equipment list of 6.8 for satisfactory electrical performance of the subassemblies of the RT-1028/APN-202 Receiver Transmitter listed below.

(1) I.F. Amplifier Assembly

(2) Pulse Assembly

(3) Modulator Assembly

(4) Power Supply Assembly

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltages of 28 ±0.5 Vdc. Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages. These characteristics shall be used when testing the equipment for compliance with the minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired that their use be investigated. When

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investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3.1 When nonrepairable subassemblies are used, a mean-time-between-failure of more than 5000 hours should be the goal.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should apecify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.7 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the

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previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

Test Harness, Radar Beacon AN/APM-231 or AN/APM-231A

Pulse Generator, HP-214A

Oscilloscope AN/USM-281A

Digital Voltmeter, HP-3439A

AC/DC Range Unit, HP-3445A

Counter, HP-5326B

Variable Attenuator, HP-355C

Variable Attenuator, HP-355D

RMS Voltmeter, HP-3400A

AC Current Probe, HP-1110A

Feedthru Termination, HP-10100B

RF Pulse Generator, Kay Electric 5070A

RF Voltmeter, Millivac MV28B

RF Sweep Oscillator, Kay Electric 866A

Power Supply, Power Designs 6050

Tee Adaptor, Series BNC, UG-274A/U

Termination, 50 ohm, Series BNC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-81922(AS) RACK, ELECTRICAL EQUIPMENT MT-4484/APM

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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